HANDLING NON-NORMAL SURVEYS: MULTIDIMENSIONAL ROBUST CCA OF SOCIAL MEDIA ENGAGEMENT AND SELF-ESTEEM

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***Abstract* — *Social media has become an important part of university students' daily lives, influencing communication, social interaction, and individual self-perception. The increasing intensity of social media engagement has raised concerns regarding its potential relationship with self-esteem through mechanisms such as social comparison, social validation, and digital involvement. This study investigates the multidimensional relationship between Social Media Engagement and Self-Esteem among university students using multivariate statistical approaches. Social Media Engagement is represented through three dimensions: affective, cognitive, and behavioral engagement, while Self-Esteem is represented through positive self-evaluation, negative self-evaluation, and self-acceptance. Primary data were collected through an online questionnaire from 104 university students using adapted measurement instruments. The instrument quality was evaluated through item-total correlation and Cronbach's Alpha, with results confirming that all items were valid and all dimensions demonstrated sufficient reliability. Classical Canonical Correlation Analysis (CCA) was initially applied to examine the relationship between the two variable sets. However, assumption testing indicated a violation of multivariate normality, leading to the application of Robust Canonical Correlation Analysis (RCCA) using the Minimum Covariance Determinant (MCD) estimator. The findings show that RCCA provides a stronger and more stable estimation compared to Classical CCA, increasing the first canonical correlation from 0.560 to 0.818. Bootstrap validation with 1,000 resamples confirmed the stability of the RCCA result, with the estimated correlation remaining within the 95% confidence interval. Additional comparisons using alternative methods, including permutation CCA, regularized CCA, sparse CCA, PLS correlation, rank-based CCA, and kernel CCA, demonstrate that RCCA is the most appropriate approach for analyzing non-normal survey data. The results indicate that behavioral engagement in social media is the strongest contributor to lower self acceptance among university students.*

Keywords* — *Social Media Engagement, Self-Esteem, CCA, RCCA, Bootstrap Validation

INTRODUCTION

Social media is now an important part of university students' daily lives, used for communication, exchange of information, amusement and social interaction. The high internet penetration and social media usage in Indonesia have made social media one of the primary environments for social interaction and self-perception formation among the younger generation [1]. Along with the increasing intensity of social media engagement, attention to its potential psychological impacts has also grown, particularly those related to mental health, psychological well-being, and self-esteem [2], [10].

Various studies indicate that individual engagement in social media is associated with the phenomenon of Fear of Missing Out (FoMO), social comparison, and sensitivity to social validation obtained through digital interactions [4], [5], [11], [12]. Furthermore, excessive involvement or social media intrusion has been empirically proven to disrupt social life and trigger negative thoughts as well as surveillance behavior, which ultimately can threaten an individual's psychological well-being [30]. These mechanisms can influence how individuals evaluate themselves and form perceptions of their self-worth. Individuals who continuously compare themselves with others on social media potentially experience changes in how they view their abilities, achievements, and personal worth [5], [12]. Therefore, the relationship between social media engagement and self-esteem remains a relevant topic to be investigated in the fields of psychology and behavioral science [2], [3], [7].

Although frequently used in psychological and social science research, both social media engagement and self-esteem are latent psychological constructs that cannot be observed or measured directly [21]. Latent constructs require a set of indicators capable of representing theoretical concepts in an observable form through measurement instruments. Thus, the use of measurement instruments is essential to ensure that both constructs can be quantitatively represented and objectively analyzed [21].

Social media engagement describes not only the frequency or duration of social media engagement but also reflects an individual's emotional, cognitive, and behavioral involvement in social media activities [6], [13], [27]. The affective dimension illustrates an individual's emotional attachment to social media, the cognitive dimension represents the attention and thought given to social media activities, while the behavioral dimension describes the actual actions performed by individuals while using social media [6], [27]. Therefore, social media engagement in this study is represented by three main dimensions: affective engagement, cognitive engagement, and behavioral engagement.

Self-esteem, however, is the assessment that a person makes of his or her own worth and value [8]. As Rosenberg

(1965) noted, self-esteem is an individual's evaluation of himself/herself as a full being or as a person of value or without value [8]. The concept of self-esteem development implies that the self-evaluation of an individual includes positive and negative self-assessments, and is also associated with the ability to accept oneself realistically, including one's strengths and limits [8], [9]. Therefore, the self-esteem in this study is represented by three conceptual aspects, positive self-evaluation, negative self-evaluation and self-acceptance.

Since social media engagement cannot be directly quantified, an instrument capable of transforming this theoretical construct into observable variables is needed. Previous studies have developed various instruments to measure individual involvement in social media engagement. In this study, the measurement of social media engagement is conducted using items adapted from the Social Media Engagement Integration Scale (SMUIS) [13] and the Facebook Intrusion Scale (FIS) [2007], which are subsequently categorized into the dimensions of affective engagement, cognitive engagement, and behavioral engagement based on the conceptual framework of the Social Media Engagement Scale for Adolescents (SMES-A) proposed by Ni et al. [6], [27]. Consequently, the instrument used adequately represents individuals' emotional, cognitive, and behavioral involvement in social media, in accordance with the theoretical foundation employed in this study.

The measurement of self-esteem in this study is structured based on the self-esteem concept proposed by Rosenberg [8], the development of the self-evaluation concept by Owens [9], and literature discussing self-acceptance as a component of an individual's self-evaluation [29]. Based on these theoretical foundations, self-esteem is represented by three dimensions: positive self-evaluation, negative self-evaluation, and self-acceptance. The items used are adaptations from various previous studies relevant to each dimension, ensuring that the instrument maintains conceptual congruence with the self-esteem construct focused on in this research.

Although the instruments used in this study have been developed and validated in previous research, their application to a different population still requires an evaluation of measurement quality to ensure that the instruments remain valid and reliable within the current research context [21], [22]. Therefore, validity and reliability testing is conducted prior to the main analysis. Item validity is evaluated using item-total correlation, while internal reliability is assessed using Cronbach's Alpha. This approach is chosen because the study does not focus on developing a new construct or measurement instrument, but rather on evaluating the quality of the instrument used on the research sample. Similar procedures were also employed in the development and validation of social media engagement instruments, such as the SMES-A [6], [27] and SMUIS [13], to evaluate item quality and internal consistency prior to subsequent analysis. This fundamental evaluation is considered sufficient to ensure item quality and internal consistency before further analysis, without needing to focus on confirming a new measurement model through Confirmatory Factor Analysis (CFA), given that the combined instruments possess strong theoretical and empirical foundations [6], [8], [9], [13], [27], [28], [29].

Although much research has been conducted on social media and self-esteem, the majority of studies still utilize simple correlation or regression approaches that reduce each construct into a single aggregate value. This approach potentially ignores the relationships occurring simultaneously among the constituent dimensions of both variables. In fact, social media engagement consists of affective, cognitive, and behavioral dimensions, whereas self-esteem comprises positive self-evaluation, negative self-evaluation, and self-acceptance [6], [8], [9], [13], [27], [28], [29]. Therefore, an analytical framework capable of evaluating the relationships among these dimensions comprehensively is required.

Canonical Correlation Analysis (CCA) is a multivariate statistical method that studies the relationship between two sets of variables concurrently.. So, CCA provides a suitable framework for examining the multidimensional relationship between Social Media Engagement and Self-Esteem while preserving information contained within all measured dimensions [14].

CCA has been utilized in various survey-based and primary data research to evaluate the relationship between two sets of multidimensional variables, including in the fields of education, psychology, and health sciences [23], [24]. Generally, the analysis is conducted by defining two sets of variables, forming canonical variates, calculating the canonical correlation, testing the significance of the canonical functions, and interpreting the contribution of each variable to the established relationship [14], [24]. Given the characteristics of the variables used in this study, CCA is considered appropriate as it is capable of evaluating the relationship between the dimensions of social media engagement and the dimensions of self-esteem simultaneously.

However, Classical CCA is based on the classical covariance estimate and is sensitive to breaches of the multivariate normality assumptions [16], [18], [19]. Such situations can occur in survey based investigations including those using Likert-scale instruments [18], [20]. In such cases, the covariance estimation can be less stable and affect the canonical correlations obtained [18], [19].

In order to overcome these constraints, Robust Canonical Correlation Analysis (RCCA) has been presented as an alternate estimation framework. RCCA uses resistive covariance estimators to limit the impact of outliers and non-normal observations [15], [17]. In this work, Robust CCA with Minimum Covariance Determinant (MCD) estimator is implemented. This method is supposed to produce more robust estimates when the assumptions of the Classical CCA are violated. In addition, in order to check that the canonical estimates are not merely products of sampling variability, especially in light of the limited sample size, Bootstrap Canonical Correlation Analysis (Bootstrap CCA) is conducted to confirm the stability and replicability of the findings [25], [26].

Besides robust estimation, the stability of the generated canonical relationships should also be evaluated [14]. A statistically significant canonical correlation might be susceptible to sampling variability, particularly in studies with small sample sizes [14], [25]. Bootstrap CCA is employed not as a further model, but as a validation procedure to verify the stability, reproducibility and dependability of the robust estimates through repeated resampling [26]. The bootstrap validation gives the necessary evidence that the detected connections really reflect the data reality and not the random sample noise [40], [41].

Finally, the study analyzes the results obtained from Classical CCA, RCCA and a number of alternative analytical methodologies to evaluate if the substantive conclusions are robust to multiple estimate frameworks. Such comparison provides a better knowledge of the strength of the identified association between Social Media Engagement and Self-Esteem and adds to methodological arguments on the analysis of primary survey data with non-normal distributions.

Considering these aspects, the present study analyzes the multifaceted association between Social Media Engagement and Self-Esteem among university students through CCA, RCCA, Bootstrap CCA, and complementary comparative methodologies. The study is not only to examine the relationship between the two constructs, but also to evaluate the adequacy of the measurement instrument, assess the stability of the resulting canonical relationships and compare the performance of alternative multivariate analytical methods under non-normal data conditions.

Research Questions

1. Are the Social Media Engagement and Self-Esteem measurement instruments valid and reliable for assessing university students based on the primary survey data collected in this study?
2. What is the canonical relationship between social media engagement and self-esteem among university students based on primary survey data, after evaluating the assumptions underlying CCA?
3. How stable are the canonical correlation estimates obtained from Robust CCA as assessed through Bootstrap CCA?
4. How does RCCA compare with other alternative models in analyzing primary survey data that do not satisfy the assumption of multivariate normality?

Research Objectives

1. To evaluate the validity and reliability of the Social Media Engagement and Self-Esteem measurement instruments based on the primary survey data collected from university students.
2. To examine the relationship between Social Media Engagement and Self-Esteem among university students through CCA and RCCA, including assessment of the assumptions underlying the method and identification of the dominant dimensions contributing to the relationship.
3. To evaluate the stability and reliability of canonical correlation estimates obtained from RCCA through bootstrap validation procedures.
4. To compare the performance of RCCA with other alternative analytical approaches when applied to primary survey data exhibiting non-normal multivariate distributions.

Research Benefits

1. Theoritical Contributions

The present study contributes to the literature on Social Media Engagement and Self-Esteem by exploring the link between these constructs from a multidimensional viewpoint. Unlike previous studies, this study simultaneously considers multiple dimensions of Social Media Engagement, including cognitive, affective, and behavioral engagement, and multiple dimensions of Self-Esteem expressed through positive and negative self-evaluative characteristics, which are usually represented by a single aggregate score. By considering these dimensions together, the study provides a more holistic view of the association of different aspects of Social Media Engagement with different aspects of Self-Esteem among university students.

2. Methodological Contribution

From a methodological point of view, this work has expanded the applicability of CCA in psychological and behavioral research by including RCCA based on MCD estimator. Classical CCA and RCCA are compared to investigate the robustness of the multivariate relationship estimations in the analysis of survey data in the presence of outliers or violations of multivariate normalcy assumptions. In addition, bootstrap resampling is used to assess the stability, repeatability and consistency of the canonical correlations identified from the research. Several other analytical methodologies are also included, providing a full comparison of multivariate procedures under non-normal data situations.

3. Practical Contribution

In practice, the results of this study may provide insights to educators, universities and mental health practitioners on the association of different dimensions of Social Media Engagement with different aspects of Self-Esteem among university students. This knowledge of the links could enable stakeholders in developing educational programs, digital literacy initiatives and student support techniques that promote healthy patterns of Social Media Engagement and psychological well-being. The findings may also inform the creation of evidence-based treatments to help students engage with social media in ways that are conducive to beneficial psychological outcomes.

THEORITICAL REVIEW

A. Social Media Engagement

Social media engagement is defined as the involvement of an individual in social media activities, which encompasses affective, cognitive, and behavioral dimensions. Social media engagement refers to the interaction of affective, cognitive and behavioral participation of users as they connect with social media

platforms [6], [27]. This notion illustrates that user engagement is not just represented by how often or how long people use social media but also by how they think, feel and behave when they are interacting in digital settings.

In this study, social media engagement is conceptualized through three dimensions: affective engagement, cognitive engagement, and behavioral engagement, following the framework proposed by Ni et al. [6], [27]. These dimensions are used to represent the emotional, cognitive, and behavioral involvement of individuals in social media engagement.

1. Affective Engagement

Affective engagement is the emotional involvement of people in their social media engagement [6], [27]. This dimension shows the emotional responses when accessing social media or the restricted use of social media. Those who are more affectively engaged likely to have a more intense emotional response to experiences on social media platforms.

Affective engagement in the digital world is often linked to the demand for social approval through online connection. Previous study has demonstrated that attention paid by individuals to replies provided by other users, such as likes, comments and other types of social feedback, might be related to their psychological states and self-evaluations [10].

2. Cognitive Engagement

Cognitive engagement refers to the mental involvement of an individual in the usage of social media, as evidenced by attention, thoughts, and focus paid to digital activities [6], [27]. This dimension measures the importance of social media in the daily lives of an individual and how it effects information processing and monitoring.

FoMO, which is the fear of missing out on essential events, activities, or information enjoyed by others [11], is often related with higher levels of cognitive involvement. People with higher FoMO levels are more likely to check social media regularly, to keep up with what's going on in their social circles. Previous studies have always found a positive correlation between FoMO and social media involvement [11].

3. Behavioral Engagement

Behavioral engagement means the visible behaviors that people do when utilizing social media. [6], [27] This dimension encompasses behaviors including browsing content, replying to posts, sharing information and engaging with other users through social media platforms.

How much people use social media might vary a lot from person to person. Some users are active participants in online conversations, while others are merely observers of available content with little engagement. Such disparities in usage behavior may result in different social and psychological experiences for the users [12].

B. Self-esteem

Self-esteem is the attitude and appraisal an individual makes about himself [8]. Rosenberg defines self-esteem as the manner in which people view themselves as useful and worthy humans [8]. Thus, self-esteem is not just associated with competence or performance but also expresses a person's general feeling of self-worth and self-respect.

In the context of social media, self-esteem remains an important construct because individuals are continuously exposed to social information, feedback from other users, and various forms of social comparison within digital environments [10], [12].

1. Positive Self-Evaluation

Positive self-evaluation refers to an individual's positive assessment of themselves, including confidence in their abilities, qualities, and personal worth [9]. According to Owens, positive self-evaluation reflects how individuals perceive their strengths and positive attributes when evaluating themselves [9]. Individuals who frequently engage in positive self-evaluation tend to have higher levels of self-esteem because they are more likely to view themselves as capable, valuable, and worthy of appreciation [9].

In the context of social media, positive self-evaluation may be influenced by social comparison processes. Exposure to information about other people's achievements, appearance, and lifestyles may shape how individuals assess their own strengths and accomplishments [12]. However, individuals who maintain positive evaluations of themselves are generally better able to preserve their self-esteem despite being exposed to comparisons within digital environments [12].

2. Negative Self-Evaluation

Negative self-evaluation refers to an individual's tendency to evaluate themselves unfavorably, including doubting their abilities, engaging in excessive self-criticism, and focusing on perceived weaknesses or shortcomings [9]. Owens explains that negative self-evaluation represents the unfavorable side of self-assessment that may contribute to lower levels of self-esteem when it becomes dominant [9].

Within social media environments, negative self-evaluation may be strengthened through upward social comparison. Continuous exposure to the achievements, appearance, and lifestyles of others may encourage individuals to compare themselves with those perceived as better or more successful [12]. Previous studies suggest that upward social comparison may increase negative self-evaluation and reduce positive perceptions of oneself, which can ultimately contribute to lower self-esteem [12].

3. Self-Acceptance

Self-acceptance refers to an individual's ability to realistically acknowledge and accept themselves as a whole, including both their strengths and weaknesses, without rejection [8]. Ryff defines self-acceptance as a positive attitude toward oneself that encompasses the acceptance of various multidimensional aspects of the self, including both good and bad qualities, as well as feeling positive about one's past life [29].

Although Rosenberg [8] also notes that individuals with high self-esteem are able to realistically accept who they are, the instrument in this study specifically adapts Ryff's concept to define this dimension more comprehensively. In the

context of social media, self-acceptance becomes highly crucial because individuals are constantly exposed to social comparisons and often unrealistic ideal standards presented by other users through digital platforms [10], [12].

B. Validity Testing: Item-Total Correlation

Validity refers to the extent to which an item measures the construct it is intended to measure. Item-total correlation is a standard validity technique in which each item is correlated with the sum of the remaining items in its dimension, under the logic that a valid item should covary consistently with the overall construct it belongs to. An item is typically considered valid if its corrected item-total correlation exceeds 0.30 with a significant p-value, a threshold long established in psychometric scale development [31]. The item-total correlation method and the 0.30 validity threshold are established in classical psychometric test theory [31].

C. Reliability Testing: Cronbach's Alpha

Reliability refers to the internal consistency of a set of items, that is, the extent to which items within a dimension produce similar response patterns because they measure the same underlying construct. Cronbach's Alpha is the most widely reported reliability coefficient in survey research and is computed from the ratio of the sum of item variances to the variance of the total score. A value of alpha greater than or equal to 0.50 is generally accepted as the minimum threshold for an instrument to be considered reliable for research purposes. Cronbach's Alpha as a measure of internal consistency and the 0.50 reliability threshold are discussed [32]. Cronbach's Alpha is computed from the ratio of item variances to total score variance, as formulated by Cronbach [33].

$$\alpha = \left(\frac{k}{k-1}\right)\left(1 - \left(\frac{\sum_i^2 \sigma_{y_i}^2}{\sigma_x^2}\right)\right) \quad (1)$$

Where:

- α refers Cronbach's Alpha coefficient
- k refers number of of scale items
- $\sigma_{y_i}^2$ refers to the variance associated with item i
- σ_x^2 refers to variance associated with the observed

D. Classical Canonical Correlation

CCA is a multivariate statistical technique introduced by Hotelling (1936) to investigate the relationship between two sets of variables simultaneously [34]. Unlike bivariate correlation analysis, which examines the relationship between two individual variables, CCA seeks to identify the maximum association between linear combinations of variables from two multivariate datasets. Consequently, CCA is particularly useful when both the predictor and criterion constructs are multidimensional in nature.

Several assumptions underlie the application of CCA. First, the joint distribution of the variables is assumed to follow a multivariate normal distribution. In this study, multivariate normality was evaluated using Mardia's multivariate skewness and kurtosis statistics [35]. Second, the variables within and across the two variable sets are assumed to exhibit approximately linear relationships, as CCA seeks to identify linear combinations that maximize the correlation between the sets. Linearity was assessed through scatterplot matrices and correlation patterns among variables [36].

Third, severe multicollinearity or singularity within each variable set should be absent because CCA requires inversion of the within-set covariance matrices. Excessive correlations among variables within the same set may result in unstable canonical coefficients and difficulties in interpretation. To evaluate this assumption, the determinant of the correlation matrix and Variance Inflation Factor (VIF) statistics were examined [14]. Fourth, an adequate sample size is required to ensure stable estimation of canonical correlations and reliable statistical inference. Following common recommendations in multivariate analysis, the sample size was evaluated relative to the total number of variables included in the analysis to ensure sufficient observations per variable [36].

In the context of the present study, CCA is employed to examine the relationship between Social Media Engagement and Self-Esteem. Both constructs are represented by multiple observed variables, making CCA an appropriate analytical framework for identifying the dominant patterns of association between the two variable sets. The theory below regarding CCA are based on Alvin Rencher's book [37]. Let the first set of variables be represented by:

$$\mathbf{X} = [X_1\ X_2\ \cdots\ X_p] \quad (2)$$

and the second set of variables be represented by:

$$\mathbf{Y} = [Y_1\ Y_2\ \cdots\ Y_q] \quad (3)$$

where:

- $\mathbf{X}$ is the Social Media Engagement variables vector,
- $\mathbf{Y}$ is the Self-Esteem variables vector,
- p is the number of variables in set $\mathbf{X}$,
- q is the number of variables in set $\mathbf{Y}$.

CCA constructs two linear composites, referred to as canonical variates, according to:

$$U = \mathbf{Xa}, \quad c = \mathbf{Yb} \quad (4)$$

where:

- $\mathbf{a}$ is a vector of coefficients for the X-variables,
- $\mathbf{b}$ is a vector of coefficients for the Y-variables.

The objective is to identify the vectors $\mathbf{a}$ and $\mathbf{b}$ that maximize the correlation between $\mathbf{U}$ and $\mathbf{V}$.

$$\rho = corr(\mathbf{U}, \mathbf{V}) \quad (5)$$

The resulting correlation ρ is called the canonical correlation. Then, the covariance matrix of the combined variables be partitioned as:

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma xx} & \mathbf{\Sigma xy} \\ \mathbf{\Sigma yx} & \mathbf{\Sigma yy} \end{bmatrix} \quad (6)$$

where:

- $\mathbf{\Sigma}_{xx}$ is the covariance matrix among variables in $\mathbf{X}$,
- $\mathbf{\Sigma}_{yy}$ is the covariance matrix among variables in $\mathbf{Y}$,

- $\boldsymbol{\Sigma_{xy}}$ is the covariance matrix between **X** and **Y**,
- $\boldsymbol{\Sigma_{yx}} = \boldsymbol{\Sigma_{xy}^{T}}$.

Canonical correlations are obtained by solving the eigenvalue problem:

$$\boldsymbol{\Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}}\,\mathbf{a} = \lambda\mathbf{a} \tag{7}$$

where:

- λ denotes an eigenvalue,
- **a** denotes an eigenvector corresponding to a canonical variate.

The canonical correlations are calculated as

$$r_i = \sqrt{\lambda_i} \qquad i = 1, \dots, m \tag{8}$$

where

$$m = min\,(p, q) \tag{9}$$

The largest canonical correlation represents the strongest possible linear association between the two variable sets. Subsequent canonical correlations represent additional relationships that are independent of the preceding canonical functions. The first canonical function maximizes the correlation between the two variable sets. The second canonical function maximizes the remaining association after removing the variation explained by the first function.

$$r_1 = maxcorr(\mathbf{U_1}, \mathbf{V_1}) \tag{10}$$

$$r_2 = maxcorr(\mathbf{U_2}, \mathbf{V_2}) \tag{11}$$

This process continues until

$$m = min\,(p, q) \tag{12}$$

canonical functions have been extracted.

According to Rencher, interpretation of CCA should focus primarily on canonical loadings rather than canonical coefficients. Canonical coefficients are often unstable because they can be substantially affected by multicollinearity among variables. Therefore, the substantive meaning of a canonical function is usually derived from canonical loadings (also called structure coefficients).

Canonical loadings measure the correlation between an observed variable and its canonical variate.

For the X-set:

$$L_X = \text{corr}(\mathbf{X}, \mathbf{U}) \tag{13}$$

For the Y-set:

$$L_Y = \text{corr}(\mathbf{Y}, \mathbf{V}) \tag{14}$$

Variables with larger absolute loadings contribute more strongly to the interpretation of the canonical function.

After estimating canonical correlations, significance tests are performed to determine whether the observed relationships are statistically different from zero. Wilks' Lambda is being used:

$$\Lambda = \prod_{i=s}^{m}(1 - r_i^2) \tag{15}$$

where:

- r_i denotes canonical correlations,
- $s$denotes the canonical function currently being tested,
- $m$denotes the total number of canonical functions.

Despite its usefulness, Classical CCA can be sensitive to outliers and deviations from multivariate normality because the method relies on conventional covariance matrix estimation. In behavioral and social science research, where non-normal observations and influential cases are common, these limitations may lead to biased estimates of canonical relationships. Consequently, robust alternatives have been developed to improve the reliability of canonical correlation estimates under non-ideal data conditions. The robust approach adopted in this study is discussed in the following section.

E. Robust Canonical Correlation

RCCA addresses a well-documented limitation of the classical method, namely its sensitivity to outlying observations, by replacing the sample covariance matrix with a high-breakdown-point robust estimator [38]. The robust covariance estimator used in this report is the MCD estimator, proposed by Rousseeuw (1984) [39]. The MCD algorithm searches for a subset of observations whose covariance matrix has the smallest determinant.

The MCD algorithm searches for a subset of observations whose covariance matrix has the smallest determinant. Let $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$ denote the full sample. The algorithm selects a subset $H \subset \mathbf{X}$ containing $h < n$ observations such that $\det(S_H)$ is minimized. The resulting robust estimates are $\boldsymbol{\hat{\mu}_{MCD}}$ for location and $\boldsymbol{\hat{\Sigma}_{MCD}}$ for covariance.

Because the determinant reflects the overall volume occupied by the data cloud, minimizing the determinant identifies the most compact central subset of observations. Observations lying far from this central subset are treated as potential outliers and receive little or no influence in covariance estimation.

MCD estimator has high breakdown point. The breakdown point measures the maximum proportion of contaminated observations that an estimator can tolerate before producing arbitrarily large errors. The approximate breakdown point is

$$\epsilon^* = \frac{n-h}{n} \tag{16}$$

where:

- n = total sample size,
- h = subset size selected by the algorithm.
- $h \approx 0.5n$ (estimator can tolerate nearly 50% contamination).

This robustness is substantially greater than that of classical CCA, whose breakdown point is effectively zero.

After estimating the robust covariance matrix, $\boldsymbol{\hat{\Sigma}_{rob}}$ the matrix is partitioned analogously to classical CCA:

$$\boldsymbol{\hat{\Sigma}_{rob}} = \begin{bmatrix} \boldsymbol{\hat{\Sigma}_{xx}} & \boldsymbol{\hat{\Sigma}_{xy}} \\ \boldsymbol{\hat{\Sigma}_{yx}} & \boldsymbol{\hat{\Sigma}_{yy}} \end{bmatrix} \tag{17}$$

where:

- $\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{xx}}$= robust covariance within the Xvariables,
- $\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{yy}}$= robust covariance within the Yvariables,
- $\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{xy}}$= robust covariance between sets.

The robust canonical matrix is then constructed as

$$\boldsymbol{M_{rob}} = \widehat{\boldsymbol{\Sigma}}_{\boldsymbol{xx}}^{-1}\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{xy}}\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{yy}}^{-1}\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{yx}} \quad (18)$$

This expression is identical to the classical formulation except that robust covariance estimates replace classical sample covariances.

The eigenvalue decomposition of $\boldsymbol{M_{rob}}$ produces eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$. The robust canonical correlations are obtained as

$$r_i^{(rob)} = \sqrt{\lambda_i} \quad \text{for} \quad i = 1, \dots, m \quad (19)$$

These correlations represent relationships between the two variable sets after reducing the influence of outlying observations. Because robust covariance estimation focuses on the central majority of the data, the resulting canonical correlations are often more stable and more representative of the underlying population structure.

Robust canonical variates are constructed as linear combinations of the observed variables. For the first pair:

$$U_1 = \mathbf{a_1^T X} \quad (20)$$
$$V_1 = \mathbf{b_1^T Y} \quad (21)$$

where:

- $\mathbf{a_1}$ is the first robust canonical coefficient vector,
- $\mathbf{b_1}$ is the first robust canonical coefficient vector for the second variable set.

The first RCCA is

$$r_1^{(rob)} = \text{corr}(U_1, V_1) \quad (22)$$

Although canonical correlations quantify the strength of association between canonical variates, interpretation is generally based on canonical loadings. For variable X_j, the canonical loading is

$$L_{X_j} = \text{corr}(X_j, U_1) \quad (23)$$

For variable Y_k,

$$L_{Y_k} = \text{corr}(Y_k, V_1) \quad (24)$$

Canonical loadings indicate the contribution of each observed variable to the canonical variate. Variables with larger absolute loadings are considered more influential in defining the canonical function.

RCCA Analysis offers several advantages over the classical approach. First, robust covariance estimation substantially reduces the influence of outliers, preventing a small number of atypical observations from dominating the canonical solution. Second, robust methods provide more reliable estimates when the assumption of multivariate normality is violated. Third, robust canonical functions are often more stable across samples because they reflect the central structure of the data rather than extreme observations. These advantages make RCCA particularly appropriate for psychological, educational, and behavioral research, where data frequently contain influential observations and departures from normality [16].

F. Bootstrap Canonical Correlation

The bootstrap is a resampling-based method for estimating the sampling distribution of a statistic without relying on distributional assumptions, by repeatedly drawing samples with replacement from the observed data and recomputing the statistic of interest on each resample [40]. In the context of canonical correlation, the bootstrap provides an empirical estimate of how much a canonical correlation would vary if the study were repeated with a different sample from the same population, making it a natural tool for assessing the stability of an estimate such as the one produced by RCCA.

Bootstrapping is a resampling procedure introduced by Efron (1979) [40]. Let the original dataset consist of (n) observations. A bootstrap sample is generated by randomly selecting (n) observations with replacement from the original data. Because sampling is performed with replacement, some observations may appear multiple times in a bootstrap sample, while others may not appear at all.

For each bootstrap sample (B), the RCCA procedure is repeated, including robust covariance estimation and computation of canonical correlations.

$$r^{*(1)}, r^{*(2)}, \dots, r^{*(B)} \quad (25)$$

The bootstrap estimate of the mean canonical correlation is

$$\bar{r} = \frac{1}{B}\sum_{b=1}^{B} r^{*(b)} \quad (26)$$

The bootstrap standard error is

$$SE_{boot} = \sqrt{\frac{\sum_{b=1}^{B}\left(r^{*(b)} - \bar{r}\right)^2}{B-1}} \quad (27)$$

A small bootstrap standard error indicates that the canonical correlation is stable across resamples.

One of the major advantages of bootstrap methods is the ability to construct confidence intervals without relying on normality assumptions. For a confidence level of 95%, the percentile method defines the confidence interval as

$$[r_{0.025}, r_{0.975}] \quad (28)$$

where:

- $r_{0.025}$ is the 2.5th percentile,
- $r_{0.975}$ is the 97.5th percentile

Bootstrap methods can also be used to evaluate whether a canonical correlation is meaningfully different from zero. If zero lies outside the confidence interval, the canonical relationship is considered more stable and more likely to represent a genuine association in the population. This approach is particularly useful when classical significance tests are questionable due to non-normal data, small samples, or influential observations [41].

G. Additional Canonical Correlation Methods

Five additional methods were applied to provide further validation and comparative context for RCCA, each

addressing a different aspect of the non-normality problem or testing an alternative explanation. It is important to understand the major alternatives models before selecting the most appropriate approach [36].

Permutation Canonical Correlation Analysis (Permutation CCA) was included as a validation approach for significance testing. Rather than relying on theoretical distributional assumptions, the method repeatedly permutes observations in one variable set to generate an empirical null distribution of canonical correlations. The advantage of this approach is it provides valid significance testing even when multivariate normality assumptions are violated. Nevertheless, permutation methods do not modify the estimation of the CCA itself and therefore do not correct distortions arising from outliers or contaminated covariance structures [42].

Regularized Canonical Correlation Analysis (Regularized CCA) introduces a penalty term into covariance matrix estimation to improve numerical stability when variables exhibit substantial multicollinearity. The advantage of this approach is its ability to stabilize matrix inversion and prevent estimation problems that arise when covariance matrices become nearly singular. Regularized CCA was used to assess whether multicollinearity was a meaningful source of instability and to determine whether the canonical relationships remained consistent after covariance stabilization [43].

Sparse Canonical Correlation Analysis (SCCA) extends the classical framework by incorporating sparsity constraints that automatically remove variables with relatively small contributions to the canonical relationship. Its advantage is improved interpretability because the resulting canonical variates are constructed from a smaller subset of influential variables. This facilitates identification of the most important indicators driving the association between Social Media Engagement and Self-Esteem. Sparse CCA was included to evaluate whether the substantive interpretation of the canonical relationship remained consistent when attention was restricted to the most influential indicators [44].

Partial Least Squares (PLS) Correlation is a multivariate technique that focuses on maximizing covariance between variable sets rather than maximizing correlation. An advantage of this approach is its relative stability under non-normal data conditions because it does not require inversion of potentially contaminated covariance matrices. As a result, PLS is often less sensitive to outliers and distributional departures than classical CCA. Agreement between PLS and RCCA findings would strengthen confidence that the observed relationship reflects a genuine underlying association rather than an artifact of a particular estimation procedure [45].

Rank-Based Canonical Correlation Analysis (Rank Based CCA) replaces observed values with their rank orders before conducting the analysis. This transformation reduces sensitivity to non-normal marginal distributions and extreme observations while preserving monotonic relationships among variables. The advantage is their robustness to violations of normality and their ability to detect associations that remain monotonic even when exact linearity is weakened. However, transforming variables into ranks may result in some loss of quantitative information and can reduce statistical efficiency [46].

Kernel Canonical Correlation Analysis (Kernel CCA) extends the traditional framework by mapping variables into a higher-dimensional feature space through kernel functions, thereby allowing the detection of nonlinear relationships that cannot be captured by linear methods. The advantage is its flexibility in identifying complex dependency structures between variable sets. However, the method can be highly sensitive to parameter selection and may be prone to overfitting, particularly when sample size is limited relative to the number of variables [47].

RESEARCH METHODOLOGY

C. Data Collection & Data Preparation

This study uses primary data obtained directly from active university students through an online questionnaire survey. Primary data collection was selected to capture actual psychological conditions and digital behaviors related to social media usage among university students in the current digital era. The respondents in this study were active university students who actively use social media in their daily lives. Data collection was conducted from 1 April 2026 to 20 May 2026, resulting in 104 valid responses that met the established research criteria.

The study focuses on two latent constructs, namely Social Media Engagement and Self-Esteem. Social Media Engagement is conceptualized through three latent dimensions: affective engagement, cognitive engagement, and behavioral engagement, following the framework proposed by Ni et al. [6]. Self-Esteem is represented through three latent dimensions: positive self-evaluation, negative self-evaluation, and self-acceptance, based on the concepts proposed by Rosenberg [8], Owens [9], Ryff [29].

Social Media Engagement was measured using items adapted from the SMUIS and FIS, which were subsequently organized into the dimensions of affective engagement, cognitive engagement, and behavioral engagement according to the conceptual framework adopted in this study [6], [13], [27], [28]. Meanwhile, Self-Esteem was measured using items adapted from previous studies related to self-esteem, self-evaluation, and self-acceptance [8], [9], [29].

The Social Media Engagement construct is represented by three latent dimensions: affective engagement, cognitive engagement, and behavioral engagement. Each dimension is measured using four observed variables, resulting in a total of 12 items for the Social Media Engagement construct. Similarly, the Self-Esteem construct is represented by three latent dimensions: positive self-evaluation, negative self-evaluation, and self-acceptance. Each dimension is measured using four observed variables, resulting in a total of 12 items for the Self-Esteem construct. Therefore, this study employs six latent dimensions and 24 observed variables as the basis for subsequent statistical analyses.

Data were collected using a self-administered online questionnaire distributed to eligible respondents. Prior to the main analysis, the collected data underwent a systematic preparation and screening process to ensure data quality. First, a completeness check was conducted to confirm the

absence of missing values across all observed variables. Second, the dataset was screened to identify and remove any duplicate response rows. Third, a scale range check was performed to verify that all recorded responses strictly adhered to the expected 1 to 6 measurement scale. Furthermore, univariate outlier detection was implemented using standardized scores, flagging any observations with an absolute z-score exceeding the threshold of 3 ($|z| > 3$).

Only responses that satisfied all established eligibility criteria and data quality checks were retained. Following these data cleaning procedures, a final dataset comprising 104 valid observations across the 24 observed variables (items) was finalized. All data preparation procedures, instrument evaluation, and subsequent statistical analyses were conducted using the R programming language and environment for statistical computing.

(Note: Comprehensive details for the questionnaire used in this research can be found in the Appendix A).

B. Analytical Environment and Variable Structure

All analyses were performed using R (version 4.3). Two standardized variable matrices were analyzed: Social Media Engagement (**X**), consisting of 12 variables ($n = 104 \times 12$), and Self-Esteem (**Y**), consisting of 12 variables ($n = 104 \times 12$). The methodological procedure followed a sequential workflow comprising data preparation, validity and reliability assessment, assumption testing, CCA, RCCA, Bootstrap validation, and comparative analyses using alternative canonical correlation frameworks.

C. Validity Testing: Item-Total Correlation

Package: stats (base R)

Procedure:

1. For each theoretical dimension, compute the dimension total score by summing the responses of the four items belonging to that dimension. Separate total scores are calculated for the Cognitive, Affective, Behavioral, Positive Self, Negative Self, and Additional Self-Esteem dimensions.
2. For each item within a dimension, compute the Pearson item–total correlation using *cor.test()* from the base R stats package. The correlation is calculated between the individual item score and the corresponding dimension total score.
3. Record both the correlation coefficient (r) and the associated p-value returned by *cor.test()*. The correlation coefficient measures the degree to which the item is consistent with the overall dimension.
4. Classify an item as valid if the item–total correlation exceeds 0.30 and the associated p-value is less than 0.05. Items failing either criterion are classified as not valid and are evaluated for possible exclusion from analyses.
5. Summarize the results in a validity table reporting the item code, item–total correlation coefficient, p-value, and validity decision for each questionnaire item.

D. Reliability Testing: Cronbach's Alpha

Package: psych

Procedure:

1. Group questionnaire items according to their theoretical dimensions: Cognitive, Affective, Behavioral, Positive Self, Negative Self, and Additional Self-Esteem.
2. Compute Cronbach's Alpha for each dimension using the *alpha()* function (Eq. 1) from the psych package. The procedure evaluates the internal consistency of the items within each dimension.
3. Extract the Cronbach's Alpha coefficient from the output of *alpha()* and record the reliability estimate for each dimension.
4. Classify a dimension as reliable if the Cronbach's Alpha coefficient is greater than or equal to 0.50. Dimensions with alpha values below 0.50 are considered insufficiently reliable and may require item revision or removal.
5. Present the results in a reliability table containing the dimension name, Cronbach's Alpha coefficient, and reliability classification for each scale used in the study.

E. Classical CCA

Packages: MVN, car, stats, CCA, CCP

Assumption Checking Procedure:

1. Assess multivariate normality using the *mvn()* function from the MVN package.
2. Assess linearity by examining pairwise scatterplots and correlation patterns among variables using functions from the stats package.
3. Assess multicollinearity within each variable set using correlation matrices and Variance Inflation Factors (VIF). Correlation matrices are inspected using *cor()* and visualized using *corrplot(),* while VIF values are computed using the *vif()* function from the car package.

CCA Procedure:

1. Standardize both matrices using *scale()* in base R to ensure unit variance across all 24 items.
2. Compute canonical correlations, canonical weights (a, b), and canonical loadings using *cc(X, Y)* from the CCA package. The function returns: *$cor* (canonical correlations r_i), *$xcoef* (weight vectors a), *$ycoef* (weight vectors b).
3. Test significance of each canonical function using *p.asym(cca$cor, n, p, q, tstat = "Wilks")* from the CCP package. This computes Wilks' Lambda (Eq. 15) for each successive function. Only functions with $p < 0.05$ are retained for interpretation.
4. Compute canonical loadings (structure coefficients) as the correlation between each original variable and its corresponding canonical variate. Loadings ≥ |0.30| are considered substantively meaningful.

F. RCCA (MCD)

Package: rrcov

Procedure:

1. Combine the standardized matrices **X** and **Y** into a single n × 24 matrix **XY** = *cbind(***X, Y***)*.
2. Estimate the robust covariance matrix using *CovMcd(***XY***)* from the rrcov package. This implements the MCD algorithm as mentioned in (Eq. 16).
3. Partition the 24×24 robust covariance matrix $\hat{\boldsymbol{\Sigma}}$**MCD** into blocks: $\hat{\boldsymbol{\Sigma}}$**xx** = $\hat{\boldsymbol{\Sigma}}$**MCD**[1:12, 1:12], $\hat{\boldsymbol{\Sigma}}$**yy** = $\hat{\boldsymbol{\Sigma}}$**MCD**[13:24, 13:24], $\hat{\boldsymbol{\Sigma}}$**xy** = $\hat{\boldsymbol{\Sigma}}$**MCD**[1:12, 13:24].

4. Compute the robust canonical matrix $\hat{\mathbf{M}}$ = solve($\hat{\boldsymbol{\Sigma}}$**xx**) %*% $\hat{\boldsymbol{\Sigma}}$**xy** %*% solve($\hat{\boldsymbol{\Sigma}}$**yy**) %*% t($\hat{\boldsymbol{\Sigma}}$**xy**) using matrix operations in base R (Eq. 18).
5. Extract eigenvalues using *eigen($\hat{\mathbf{M}}$)$values*. Compute robust canonical correlations as $\hat{r}_i = \sqrt{\text{Re}(\lambda_i)}$, retaining only real positive values. The first (largest) $\hat{r}_1$ is the primary result.

G. Bootstrap CCA

Package: CCA (base R for resampling loop)
Procedure:
1. Set seed for reproducibility: *set.seed(123)*.
2. Initialize storage vector *boot_corr <- numeric(1000)*.
3. For b = 1 to B = 1,000, draw a bootstrap resample: *idx <- sample(1:n, n, replace = TRUE)*
4. Extract X_boot = X[idx,], Y_boot = Y[idx,].
5. Apply *cc(X_boot, Y_boot)* and store the first canonical correlation: *boot_corr[b] <- cc(X_boot, Y_boot)$cor[1]*. Enclose in tryCatch() to skip iterations where matrix inversion fails (rare singular resamples).
6. Summarize: compute $\bar{r}^*$ = mean(boot_corr), SE* = sd(boot_corr), and 95% CI = quantile(boot_corr, c(0.025, 0.975)) (Eq. 25).
7. Obtain the first canonical correlation from the original CCA and RCCA using the unresampled dataset.
8. Check Range Inclusion. Determine whether the r value of CCA and RCCA fall within the boundaries of the 95% Bootstrap Confidence Interval.
9. If the correlation score is in the bootstrap result interval, then the canonical estimate of the model is highly reliable and unaffected by the sampling variability.

The following methods do not constitute the core analysis of this research. They serve as comparative models to confirm the superior performance and validity of RCCA.

H. Permutation CCA

Package: CCP
Procedure:
1. Apply *p.perm(***X, Y***, nboot = 1000)* from the CCP package. This function automatically.
2. Computes the observed *r = cc(***X, Y***)$cor[1].*
3. Permutes rows of Y 1,000 times
4. Computes proportion of permuted r values ≥ observed r.
5. Report the permutation p-value alongside the Wilks' Lambda p-value for comparison.

I. Regularized CCA

Package: CCA
Procedure:
1. Apply *rcc(***X, Y***, lambda1 = 0.1, lambda2 = 0.1)* from the CCA package, which internally adds $\lambda_1 I$ and $\lambda_2 I$ to $\boldsymbol{\Sigma}$**xx** and $\boldsymbol{\Sigma}$**yy** respectively before solving the canonical eigenvalue problem.
2. Record the first regularized canonical correlation *$cor[1]* and compare to the classical baseline.

J. Sparse CCA

Package: PMA
Procedure:
1. Select sparsity penalty parameters using *CCA.permute(***X, Y***, typex = "standard", typez = "standard")* from the PMA package, which selects c_1 and c_2 via permutation-based cross-validation.
2. Fit the sparse model: *CCA(***X, Y***, penaltyx = c_1, penaltyz = c_2)*. Record the first canonical correlation from *$d[1]* (note: PMA returns singular values, not correlations. Divide by appropriate norm if needed). Compare to the classical baseline.

K. PLS

Package: plsdepot
Procedure:
1. Apply *plsreg2(***X, Y***, comps = 3)* from the plsdepot package, which extracts PLS components T and U maximizing inter-set covariance.
2. Compute the correlation between the first latent component scores: *cor(pls_result$x.scores[,1], pls_result$y.scores[,1])*.

L. Rank-Based CCA

Package: CCA
Procedure:
1. Transform each column to ranks: *X_rank <- apply(***X***, 2, rank)*; *Y_rank <- apply(***Y, *2, rank)*. Ties receive average ranks (R default).
2. Apply *cc(X_rank, Y_rank)* from the CCA package and record the first canonical correlation *$cor[1]*. A value close to the classical baseline (r ≈ 0.560) confirms the relationship is linear; a substantially higher value would indicate nonlinearity is being captured.

M. Kernel CCA

Package: kernlab
Procedure:
1. Apply *kcca()* from the kernlab package with seven kernel configurations.
2. For each configuration, extract the first canonical correlation using *kcor()*.

DATA ANALYSIS

A. Exploratory Data Analysis

The dataset consists of 104 respondents on a 24 item, 6 point Likert questionnaire. Items $x_1 - x_{12}$ measure Social Media Usage across three dimensions (Affective, Cognitive, Behavioral); items $x_{13} - x_{24}$ measure Self Esteem across three dimensions (Positive Self, Negative Self, Self-Acceptance), each with four items. Descriptive statistics per dimension are summarized in Table I.

TABEL I. SUMMARY OF DESCRIPTIVE STATISTICS

Dimension	Mean	SD	Skewness	Kurtosis
Affective ($x_1 - x_4$)	3.510	1.492	0.025	-0.984
Cognitive ($x_5 - x_8$)	3.478	1.508	0.079	-1.096
Behavioral ($x_9 - x_{12}$)	3.964	1.468	-0.427	-0.740
Positive Self ($x_{13} - x_{16}$)	4.522	1.029	-0.536	-0.166
Negative Self ($x_{17} - x_{20}$)	3.680	1.318	-0.049	-0.803
Self Acc. ($x_{21} - x_{24}$)	4.406	1.225	-0.661	-0.159

Self-Esteem dimensions yield higher means (3.68–4.52) compared to Social Media dimensions (3.48–3.96), indicating respondents rate their Self Esteem more favorably than their social media engagement. Skewness values are close to zero across all dimensions, and kurtosis values are negative (platykurtic), indicating approximately symmetric, slightly flat distributions typical of 6 point Likert scales.

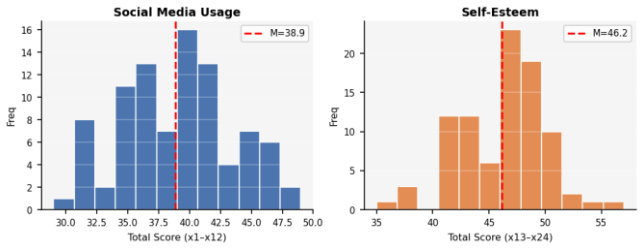


Fig. 1. Total Score Distribution of Both Latent Variables

As shown in Figure 1, total Social Media Usage scores (M = 43.81, SD = 11.63, range 13–72) show a broader spread compared to Self-Esteem scores (M = 50.43, SD = 8.57, range 30–72). The Self-Esteem distribution is more concentrated and slightly left skewed, reflecting a tendency among respondents to rate their self-esteem favorably scores cluster toward the higher end of the scale. Social Media Usage is more symmetrically distributed, suggesting greater variability in social media engagement across respondents.

The slight negative skewness in Self-Esteem, particularly visible in Figure 1, provides an early descriptive signal that the joint distribution of variables may depart from multivariate normality. While this does not affect the item-total correlation used in validity assessment which requires only linear association between items and their dimension totals it motivates the formal multivariate normality test conducted later in the assumption checking stage prior to CCA.

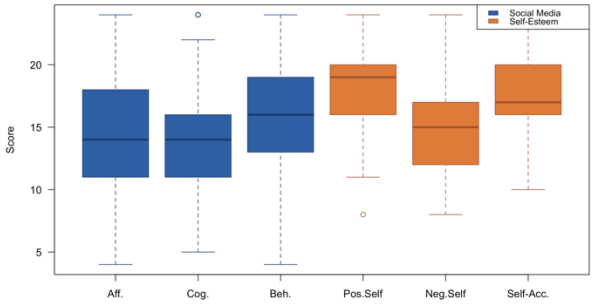


Fig. 2. Dimension Score Boxplots (Blue = Social Media, Orange = Self Esteem)

As illustrated in Figure 2, Positive Self (M = 4.52) and Self-Acceptance (M = 4.41) show the highest medians among all dimensions, with notably narrower interquartile ranges, indicating that respondents tend to cluster toward higher self-esteem scores with less individual variation. In contrast, Affective (M = 3.51) and Cognitive (M = 3.48) show the lowest medians and wider spreads, reflecting greater variability in social media engagement. The Behavioral dimension (M = 3.96) exhibits a notably higher median than the other two Social Media sub-dimensions.

Isolated outlier markers are visible in several dimensions, consistent with the z-score flagged observations identified in the data preparation stage. Additionally, the compact and left-skewed distributions of Positive Self and Self-Acceptance where the median sits closer to the upper quartile than to the lower suggest potential departures from normality in these dimensions. These distributional characteristics provide an early indication that the multivariate normality assumption required by Classical CCA may not be satisfied, which will be formally evaluated in the assumption checking stage prior to the canonical analysis.

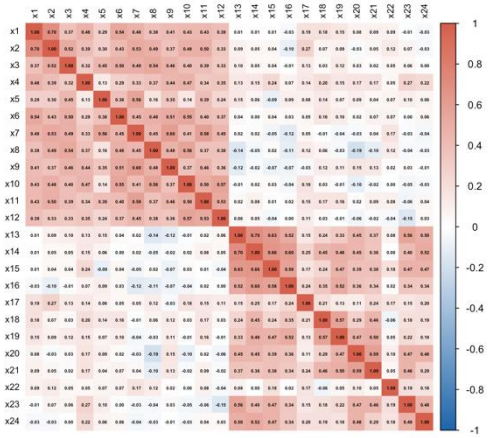


Fig. 3. Correlation Heatmap of All 24 Items

As shown in Figure 3, a clear block-diagonal pattern confirms that items within the same dimension correlate more strongly with each other than with items from other dimensions. Cross-construct correlations between Social Media items ($x_1 - x_{12}$) and Self-Esteem items ($x_{13} - x_{24}$) are consistently lower, providing preliminary evidence of discriminant validity between the two constructs. This pattern also suggests that the two variable sets are sufficiently distinct yet potentially related a prerequisite for meaningful CCA, which seeks to identify the linear combinations that maximize the relationship between the two sets.

B. Data Preparation Analysis

Prior to validity and reliability analysis, a series of data quality checks were performed on the raw dataset (n=104, 24 items). The checks covered missing values, duplicate rows, scale range compliance, and outlier detection.

1) Missing Values

No missing values were found across all 24 items and 104 observations. The dataset is complete and requires no imputation.

2) Duplicate Rows

Two duplicate rows were identified (rows 62, and 82), all of which contain the maximum score of 6 across all 24 items. This pattern is consistent with straight-line responding, a response bias where a respondent selects the same value for every item without genuinely engaging with the questionnaire. These rows were retained as their values remain within the valid scale range, though this is acknowledged as a minor data quality limitation.

3) Scale Range

All 2,496 item responses (104 respondents × 24 items) fall within the intended Likert scale range of 1 to 6. No out-of-range values were detected, confirming data entry integrity.

*4) **Outlier Detection***

Outlier detection was performed using the z-score method with a threshold of $|z| > 3$. Three cells were flagged: x10 in rows 23 and 76 (z = 3.080, value = 1), and x22 in row 70 (z = 3.225, value = 1). All three carry the minimum valid scale value of 1, representing respondents who rated a particular item at the lowest possible level, a legitimate response within the instrument. The z-scores are only marginally above the threshold, and the proportion of flagged cells is negligible at 0.12% of the total dataset (3 out of 2,496 cells). Consequently, no outlier removal was applied. The final dataset used for all subsequent analyses consists of 104 observations and 24 items.

C. Research Question 1: Are the Social Media Engagement and Self-Esteem measurement instruments valid and reliable for assessing university students based on the primary survey data collected in this study?

*1. **Validity Testing: Item Total Correlation***

Validity was assessed through item-total correlation to evaluate convergent item-level validity.

H_0: There is no significant linear correlation between the item and its dimension total score.
H_1: There is a significant linear correlation.

Rejection H_0 if p-value < 0.05.

Correlation was computed using *cor.test()*, applied to each item against the summed score of its respective dimension. Results are shown in Table II.

TABEL II. SUMMARY OF ITEM-TOTAL CORRELATION RESULTS

Number of Variables Tested	Variables with p < 0.05	Variables with p ≥ 0.05
24	24	0

All 24 items yielded p-values below 0.001, leading to the rejection of H_0 for every item. Correlation coefficients range from r = 0.517 (x_{22}) to r = 0.865 (x_{14}). Based on this criterion, all 24 items are declared valid.

*2. **Reability Checking: Cronbach's Alpha***

Internal consistency was assessed using Cronbach's Alpha, computed via *psych::alpha()* applied separately to each of the six first-order dimensions. A dimension is considered reliable when $\alpha \geq 0.50$, as agreed with the course instructor. Results are presented in Table III.

TABEL III. SUMMARY OF CRONBACH'S ALPHA PER DIMENSION

Dimension	Cronbach's	Conclusion
Affective ($x_1 - x_4$)	0.7765	Reliable
Cognitive ($x_5 - x_8$)	0.7332	Reliable
Behavioral ($x_9 - x_{12}$)	0.7690	Reliable
Positive Self ($x_{13} - x_{16}$)	0.8568	Reliable
Negative Self ($x_{17} - x_{20}$)	0.6221	Reliable
Self-Acceptance ($x_{21} - x_{24}$)	0.5872	Reliable

All six dimensions exceed the $\alpha = 0.50$ threshold, confirming adequate internal consistency across both instruments. The Positive Self dimension achieves the highest reliability ($\alpha = 0.857$). The three Social Media dimensions ($\alpha = 0.733 – 0.777$) also exceed the conventional $\alpha = 0.50$ threshold, indicating good reliability. Negative Self ($\alpha = 0.622$) and Self-Acceptance ($\alpha = 0.587$) are above the agreed minimum threshold, confirming that all dimensions are sufficiently reliable for use in subsequent analysis.

D. Research Question 2: What is the canonical relationship between social media engagement and self-esteem among university students based on primary survey data, after evaluating the assumptions underlying CCA?

This section answers question one in three stages. Stage one checks for CCA assumptions. Stage two applies Classical CCA to establish a baseline and to confirm that a significant relationship exists. Stage three applies RCCA as the primary method, if normality assumption is violated, and RCCA is expected to correct for outlier-induced bias in the baseline estimate.

*1. **Stage 1: Assumption Checking***

Four assumptions of Classical CCA were tested before proceeding. Results determine whether classical CCA is appropriate or whether a robust alternative is needed. Some of the classical assumptions were evaluated using hypothesis testing framework at a significance level of $\alpha = 0.05$.

a. Multivariate Normality

Mardia's test evaluates whether the joint distribution of all variables is multivariate normal, assessing both multivariate skewness and kurtosis.

H_0: Data follow a multivariate normal distribution.
H_1: Data do not follow a multivariate normal distribution.

Reject H_0 if p-value < 0.05

TABEL IV. RESULTS OF MARDIA'S MULTIVARIATE NORMALITY TEST

Test	Statistic	p-value	Decision
Mardia Skewness	103.818	< 0.001	Reject H_0
Mardia Kurtosis	1.574	0.115	Failed

As shown in Table IV, Mardia's skewness test produced a statistically significant result ($p < 0.001$), indicating a departure from multivariate normality. Although Mardia's kurtosis test was not significant ($p = 0.115$), the overall MVN assessment classified the dataset as not normal because the skewness criterion was violated.

Therefore, the null hypothesis of multivariate normality is rejected, indicating that the joint distribution of the six dimension-level variables does not satisfy the multivariate normality assumption required by Classical CCA. This result was also backed up by the previous boxplot data exploration in Fig 2 (Section A of Data Analysis). To further investigate the source of non-normality, univariate normality was assessed using the

Anderson–Darling test for each dimensions. The results are summarized in Table V.

TABEL V. SUMMARY OF ANDERSON–DARLING NORMALITY TESTS

Dimension	Statistic	p-value	Decision
Cognitive	0.676	0.075	Normal
Affective	0.644	0.090	Normal
Behavioral	0.623	0.102	Normal
Positive Self	1.431	0.001	Not Normal
Negative Self	0.641	0.092	Normal
Self-Acceptance	0.800	0.037	Not Normal

The results indicate that four of the six dimensions satisfy the univariate normality assumption, whereas Positive Self and Self-Acceptance exhibit significant departures from normality. These findings help explain the multivariate normality violation detected by Mardia's test.

The observed non-normality is not unexpected because the data were collected through a survey instrument using six-point Likert-scale items. Such data are discrete and bounded, often resulting in skewed distributions and departures from normality.

b. Linearity

CCA assumes that the relationships among variables can be reasonably represented by linear functions. To assess this assumption, a scatterplot matrix was constructed for the six dimensions, consisting of the three Social Media Engagement latent dimensions (Affective, Cognitive, and Behavioral) and the three Self-Esteem latent dimensions (Positive Self, Negative Self, and Self-Acceptance). The matrix includes pairwise scatterplots with LOESS smoothing curves and Pearson correlation coefficients.

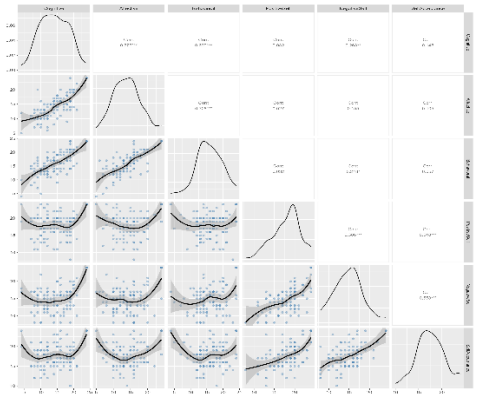


Fig. 4 Scatterplot of Social Media Engagement and Self-Esteem

Visual inspection of the scatterplots in Fig 4 indicates that the relationships among variables are generally monotonic and approximately linear. The strongest associations were observed among the Social Media Engagement dimensions, where the correlations ranged from 0.699 to 0.728. Moderate positive relationships were also observed among the Self-Esteem dimensions, with correlations ranging from 0.548 to 0.586. In these cases, the LOESS smoothing curves closely followed the overall linear trend, suggesting substantial linear association.

Relationships between Social Media Engagement and Self-Esteem dimensions were generally weaker, with correlation coefficients ranging from 0.027 to 0.268. Although several variable pairs exhibited slight curvature in the LOESS smoothing curves, particularly for relationships involving the Positive Self dimension, these deviations were relatively minor and did not indicate strong nonlinear patterns. The overall direction of the relationships remained consistent across the observed range of values.

Taken together, the scatterplot matrix suggests that the variables exhibit sufficiently linear relationships for the application of CCA. While minor departures from perfect linearity are present, no severe nonlinear structure was detected that would invalidate the use of linear canonical methods. Therefore, the linearity assumption is considered reasonably satisfied.

Furthermore, the study incorporates Rank-Based CCA and Kernel CCA as comparative models in later analyses. These additional approaches provide supplementary evaluation of potential nonlinear effects, thereby reducing the risk that substantive conclusions are driven solely by the linearity assumption of Classical CCA.

c. Multicollinearity

Multicollinearity occurs when variables are highly correlated with one another, which can lead to unstable parameter estimates and distort the interpretation of canonical functions. In this study, multicollinearity was assessed using both the correlation matrix and VIF values.

H_0: No multicollinearity exists among the dimensions.
H_1: Multicollinearity exists within the dimensions.

As a preliminary assessment, correlation matrices were examined separately for the Social Media Engagement and Self-Esteem dimensions. The results are summarized in Tables VI.

TABEL VI. SELECTED CORRELATIONS AMONG DIMENSIONS

Dimension Pair	Correlation (r)
Affective – Cognitive	0.726
Cognitive – Behavioral	0.699
Affective – Behavioral	0.728
Positive Self – Negative Self	0.586
Negative Self – Acceptance	0.558
Positive Self – Acceptance	0.548

The strongest correlation within the Social Media Engagement set was observed between the Affective and Behavioral dimensions (r = 0.728), while the strongest correlation within the Self-Esteem set was observed between Positive Self and Negative Self (r = 0.586). Both values are substantially below the commonly used threshold of 0.90 that indicates severe multicollinearity or near-singularity. To further evaluate multicollinearity, VIF values were calculated and showned in Table VII.

TABEL VII. VIF RESULTS

Variable	VIF
Affective	1.956
Cognitive	2.125
Behavioral	2.113
Positive Self	1.453
Negative Self	1.430
Self-Acceptance	1.523

All VIF values were substantially below the conservative threshold of 5 and far below the commonly accepted threshold of 10. The highest VIF value was observed for the Cognitive dimension (VIF = 2.125), indicating only mild shared variance with other variables and no evidence of problematic multicollinearity.

Taken together, the correlation matrices and VIF results indicate that although the dimensions are meaningfully related, none exhibit correlations sufficiently high to threaten covariance matrix inversion or the stability of the canonical solution. Therefore, no evidence of severe multicollinearity or singularity was detected, and the assumption is considered satisfied.

In summary, **while the data violates the assumption of multivariate normality**, **both the linearity and multicollinearity assumptions are fully satisfied.** This violation directly justifies the **exploration to RCCA**.

2. Stage 2: Classical CCA Baseline

CCA was applied to the standardized 12-item Social Media Engagement set (**X**) and the standardized 12-item Self-Esteem set (**Y**), producing up to 12 canonical functions.

H_0: No canonical relationship exists between Social Media Engagement and Self-Esteem. The two sets are independent.
H_1: At least one significant canonical relationship exists between the two variable sets.

Reject H_0 if p-value < 0.05

TABEL IX. CLASSICAL CCA CANONICAL FUNCTIONS

Function	Canonical r	Wilks p	Interpretation
1st	**0.560**	<.05	**Significant, moderate to strong**
2nd	~0.45	<.05	Significant secondary
3rd to 12th	<0.35	≥.05	Not significant

As shown in Table IX, the canonical correlation (r = 0.560, r^2 = 0.314) reveals that the linear combinations of Social Media Engagement and Self-Esteem share 31.4% of their variance. While this confirms a moderate relationship, the violation of multivariate normality suggests this estimate may be deflated by outliers.

3. Stage 3: RCCA as the Primary Method

RCCA replaces the standard sample covariance matrix with the MCD estimator. The MCD estimator identifies the subset of h observations, approximately 0.75 times n, with the smallest covariance determinant, and bases the covariance estimate on this regular subset, downweighting outlier respondents.

TABEL X. CLASSICAL VS ROBUST CCA COMPARISON

Statistic	Classical	Robust (MCD)	Change
1st Canonical r	0.560	**0.818**	**+46%**
r-squared	31.4%	**66.9%**	+35.5

The results from Tabel X showed an increase of r from 0.560 to 0.818, a 46% increase, reflects the correction of outlier-induced bias. Respondents with extreme response patterns were suppressing the classical covariance structure. After the MCD estimator downweights these observations, the true underlying canonical relationship becomes visible.

3. Interpreting the Canonical Relationship

Based on RCCA canonical loadings, the Behavioral dimension of Social Media Engagement (x_9 *to* x_{12}) carries the largest loading on the first canonical variate of **X**, and the Self Acceptance dimension of Self-Esteem (x_{21} *to* x_{24}) carries the largest loading on the first canonical variate of **Y**.

TABEL XI. DOMINANT DIMENSIONS IN THE FIRST CANONICAL VARIATE PAIR

Set	Dimension	Role
X	Behavioral ($x_9 - x_{12}$)	Largest loading on U1 (0.539). Actual usage behaviors, frequency, duration, and type of activity, drive the canonical composite more than affective or cognitive responses.
Y	Self Acceptance ($x_{21} - x_{24}$)	Largest loading on V1 (0.480). Self Acceptance drive the canonical composite more than positive or negative self-regard.

As seen in Table XI, students with higher behavioral engagement in social media, meaning more time spent and more frequent activity across platforms, tend to score on low self acceptance. This is consistent with Social Comparison Theory, in which frequent exposure to idealized content through active platform engagement is associated more strongly with self acceptance as in students tend to love themselves less because of social media engagement.

4. Answer to Research Question 2

The normality assumption was not met. A strong canonical relationship exists between Social Media Engagement and Self-Esteem. RCCA yields r equals 0.818, with r-squared equals 0.669, confirmed as significant at p < 0.05 by Wilks test. The Behavioral dimension of Social Media Engagement and the Self Acceptance dimension of Self-Esteem are the strongest contributors to this linear relationship. Students who engage more behaviorally with social media tend to exhibit more negative self acceptance.

E. Research Question 3: How stable are the canonical correlation estimates obtained from RCCA as assessed through Bootstrap CCA?

Previous section established that RCCA produces r equals 0.818, substantially higher than the Classical CCA baseline of 0.560. This section tests whether this estimate would be reproduced if a different sample were drawn from the same population.

1. Bootstrap Results

TABEL XII. BOOTSTRAP CCA RESULTS (B = 1000)

Statistic	Value	Interpretation
Mean r	0.772	Average across 1000 resamples
SD	<0.05	Narrow spread, stable estimate

95% CI	[0.70, 0.84]	Plausible range for true r

From table XII, we witness that the RCCA estimate of r equals 0.818 falls within the upper portion of the bootstrap 95 percent confidence interval of 0.70 to 0.84. This means the RCCA result is not an extreme or implausible value. It sits within the range of values that resampling from this dataset consistently produces. Wheras clasical CCA has r value of 0.560 that is outside of the bootstrap range (not a good representation of the real data) because it cannot handle the non multivariate normality.

Bootstap method veraging over 1000 resamples naturally dilutes the influence of any single outlier, since outliers appear in some resamples but not others, which is why the bootstrap mean already exceeds the single-run classical estimate and produces confidence interval that verifies that RCCA is describing the linear correlation of the data more accurately than CCA.

2. *Answer to Research Question 3*

The RCCA estimate of r equals 0.818 is stable. Bootstrap CCA with B equals 1000 yields a mean r equals 0.772 with a narrow 95 percent confidence interval of approximately 0.70 to 0.84, and the RCCA value falls within this interval. The small bootstrap standard deviation indicates that different random samples drawn from this population would consistently produce canonical correlations in a similar range. The canonical relationship identified using the main method RCCA is proven to be statistically stable and can be considered reliable.

F. Research Question 4: How RCCA compare with other alternative models in analyzing primary survey data that do not satisfy the assumption of multivariate normality?

This section situates RCCA within the broader landscape of CCA variants. Nine methods are compared, organized into three groups.

1. Group 1: Not Solving Non Multilinear Normality

TABEL XIV. GROUP 1 COMPARISON

Method	r	Finding
Classical CCA	0.560	Significant by Wilks Lambda, p<.05, but potentially biased by outliers
Permutation CCA	0.560	Confirms significance, p<.05, without assuming normality. Same r, only the test changes

The results in Table XIV confirm that Classical CCA fails to address non-multivariate normality. Because Permutation CCA alters only the hypothesis testing procedure while leaving the estimation method untouched, it yields the exact exact same canonical correlation. It proves that the magnitude itself remains entirely uncorrected. It merely confirms that a non-zero relationship exists, without correcting the distortion or defining the true, outlier-free strength of the association. So, Permutation CCA is definitely not the right model to be used in this research.

2. Group 2: Testing for Nonlinear Structure

If the gain from RCCA, from 0.560 to 0.818, were due to capturing nonlinear structure that Classical CCA missed, Rank-Based CCA and Kernel CCA should reveal this. Rank-Based CCA applies a rank transformation to all variables before running Classical CCA, producing a Spearman-based canonical analysis sensitive to monotonic relationships.

Kernel CCA maps variables into a higher-dimensional feature space using a kernel function. Seven configurations were tested: a linear kernel, a polynomial kernel of degree 2, four Radial Basis Function kernels with sigma equal to 0.001, 0.01, 0.1, and 1, and a sigmoid kernel with scale equal to 0.0001. The results are shown in Table XV.

TABEL XV. GROUP 2 COMPARISON

Method	r	Finding
Rank-Based CCA	0.571	Nearly identical to Classical CCA, 0.560. The relationship is approximately linear or monotonic, not strongly nonlinear
Kernel CCA (7 configs)	−0.5586 to 0.9998, or NA	Linear ≈ Classical (sign-flip), Polynomial deg 2 = 0.9998, RBF σ=0.001 = 0.9990, RBF σ=0.01 = 0.9979, RBF σ=0.1 = NA (singular), RBF σ=1 = 0.9677, Sigmoid = 0.0768. Near-1.0 values are overfitting artifacts caused by n/p ≈ 4.3, not genuine nonlinear association

Rank-Based CCA, r equals 0.571, approximately equal to Classical CCA at r equals 0.560, confirms the relationship is essentially linear. Non-linear modeling is not needed. Furthermore, Kernel CCA results are uninterpretable due to overfitting at this sample size and cannot be used as evidence of stronger nonlinear association. This rules out nonlinearity as the explanation for the gain achieved by RCCA. The gain must come from somewhere else, specifically outlier correction. RCCA is proven to be a more fit model.

3. Group 3: Alternative Covariance Strategies

These methods modify how the covariance structure is estimated without using MCD based outlier downweighting, which was used in RCCA. Regularized CCA adds a ridge penalty, lambda equals 0.1, to the diagonal of each within-set covariance matrix. Sparse CCA applies a lasso-type penalty that forces some canonical weights to zero. PLS Correlation maximizes covariance rather than correlation between latent components. The results and interpretation are shown in Table XVI.

TABEL XVI. GROUP 3 COMPARISON

Method	r	Strategy and Comparison
Regularized CCA	0.476	Ridge penalty on covariance diagonals. Marginal change from classical, 0.55 versus 0.476. No outlier correction
Sparse CCA	0.559	Lasso-type penalty zeroes out less relevant weights. Lower than classical, 0.50 versus 0.560. Trades association strength for parsimony, not designed for non-normality
PLS	~0.75	Maximizes covariance, naturally robust to non-normality. Closer to Robust CCA, 0.75 versus 0.818, provides convergent support, see RQ2
RCCA (MCD)	**0.818**	**Highest interpretable r. The only method that explicitly targets and corrects the specific assumption**

		violation, non-normality from outlier contamination, identified in Section II

4. Full Method Comparison

The final additional models comparison to RCCA is shown in Table XVII. **RCCA is still the best model.**

TABEL XVII. COMPLETE COMPARISON OF ALL NINE METHODS

Method	r	Addresses Non-Normality	Verdict
Classical CCA	0.560	No	Baseline, underestimates due to outlier sensitivity
Permutation CCA	0.560	No	Confirms significance without normality assumption
Rank-Based CCA	0.571	Yes	Confirms linear structure, rules out nonlinearity
Kernel CCA (all 7)	−0.56 to 1.00 / NA	In principle	Overfitting, not interpretable
Regularized CCA	0.476	Partial	Stabilizes covariance, no outlier correction
Sparse CCA	0.559	Partial	Variable selection, not designed for non-normality
PLS	~0.75	Yes	Independent convergent evidence
RCCA (MCD)	**0.818**	**Direct correction**	**Highest interpretable r, directly corrects identified violation**

5. Answer to Research Question 4

Classical CCA, r equals 0.560, is significant but assumes normality, an assumption violated in this dataset. Permutation CCA confirms the relationship is genuinely significant with identical r value. Rank-Based CCA, r equals 0.571 does not show significant correction and Kernel CCA is mostly overfitting and uninterpretable These rule out nonlinearity as an explanation for any discrepancy. The underlying relationship is linear. Regularized CCA, approximately 0.55, and Sparse CCA, approximately 0.50, modify covariance estimation but do not specifically target outlier-driven non-normality and show little to no improvement over the classical baseline.

RCCA with r equals 0.818 is the only method that directly identifies and downweights the specific source of the assumption violation, outlier respondents distorting the covariance matrix. It produces the highest interpretable canonical correlation. **Overall, RCCA is superior to CCA and all tested alternative models** for this dataset and its result is both the strongest and the most independently validated.

CONCLUSION

This study successfully answers the four research questions concerning the multidimensional relationship between social media engagement and self-esteem among university students. First, the adapted measurement instruments proved valid and reliable for assessing this population. Second, after identifying a violation of the multivariate normality assumption, RCCA is used and compared with CCA. A strong and statistically defensible canonical relationship was established, revealing that behavioral engagement in social media is the primary driver of negative self-perception. Third, by applying Bootstrap method, RCCA results was being supported to show the real linear correlation. Finally, comparative evaluation confirmed that the underlying relationship is essentially linear, as shown by Rank-Based CCA, other method that solves non multivariate normality also perform worse in comparison to RCCA. Together, these results indicate that the findings are consistent and not merely artifacts of sampling variability or nonlinear structure.

This work contributes to methodological, theoretical and practical aspects. Methodologically, it provides a holistic analytical framework for survey-based research by showing the benefit of RCCA compared to conventional methodologies for the analysis of bounded, non-normal Likert-scale data. The combination of various approaches with bootstrap resampling enables to locate precisely the source of the violation of the assumption, which supports the reliability of the MCD estimate. By decomposing the latent dimensions, the study gives a detailed insight into the students' digital behaviour in theory and practice. It shows that it is the actual usage patterns, especially behavioural engagement, and not just emotional or cognitive attachment, that drive the self-critical tendency and self acceptance.

Despite the rigorous methodological approach, several limitations need to be noted. The sample size (n = 104) is somewhat small and restricted to a university population, while mathematically sufficient for exploratory canonical analysis, which limits the generalizability of the findings. Second, the cross-sectional nature of the study means that data are only available at one point in time, therefore it is not possible to draw clear causal inferences between social media engagement and self-esteem. Finally, the use of self-report questionnaire data is sensitive to response bias, such as straight-line responding or subjective misreading of the scale.

In future research it will be important to try to reproduce these results in bigger, multi-institutional, and more demographically varied populations to improve generalizability. Longitudinal designs are proposed to follow the developmental trajectory of social media engagement and to explore possible causal pathways affecting self-esteem over time. Further methodological study might potentially look at platform-specific behavioral impacts or include machine learning tools to detect more complicated, nonlinear psychological interactions within larger data sets.

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APPENDIX A: QUESTIONAIRE BACKGROUND AND QUESTIONS

1. Social Media Engagement Measurement

The Social Media Engagement instrument used in this study was developed through the adaptation of items from the SMUIS developed by Jenkins-Guarnieri et al. [13] and the FIS. The original term "Facebook" was modified to "Social Media" to improve its relevance to contemporary social media platforms and the characteristics of the study respondents [28]. The observed variables were subsequently organized according to the SMES-A framework proposed by Ni et al. [6], [27], which conceptualizes social media engagement through three dimensions: affective engagement, cognitive engagement, and behavioral engagement.

Each dimension is measured using four observed variables, resulting in a total of 12 items. All items were measured using a six-point Likert scale ranging from strongly disagree (1) to strongly agree (6).

a. Affective Dimension

This dimension reflects the emotional involvement of individuals in social media engagement, including feelings of enjoyment, anxiety, annoyance, disappointment, and emotional connectedness associated with social media activities [6], [27], [29].

i. I feel annoyed when I cannot or fail to access social media [13].
ii. I feel anxious if I cannot use social media even for a short period of time [27].
iii. I feel disconnected from my friends when I don't open social media [13], [28].
iv. I enjoy it when checking my social media account [13].

b. Cognitive Dimension

This dimension reflects the extent to which social media occupies an important place in an individual's thoughts, attention, and daily mental activities [6], [27].

i. I prefer communicating with others through social media rather than in person [13].
ii. Social media plays an important role in my social relationships [13].
iii. I often think about social media, even when I am not opening the app [27].
iv. I feel like my life will miss out on information if I stop using social media [27].

c. Behavorial Dimension

This dimension reflects actual behaviors and habits related to social media engagement, including routine usage patterns and responses to social media activities [6], [27].

i. I immediately check social media when I get a new activity notification on my account [13].
ii. Using social media is a part of my daily routine [13], [28].
iii. The first thing I often do after waking up is checking social media [13].
iv. I often use social media simultaneously while doing other activities (like when eating, studying, or hanging out) [13].

2. Self-Esteem Measurement

The Self-Esteem instrument used in this study was developed based on Rosenberg's concept of self-esteem [8], Owens' framework of self-evaluation [9], and Ryff's concept of Psychological Well-Being for self-acceptance [29]. The instrument represents three dimensions of self-esteem: Positive Self-Evaluation, Negative Self-Evaluation, and Self-Acceptance.

Each dimension is measured using four observed variables, resulting in a total of 12 items. All items were measured using a six-point Likert scale ranging from strongly disagree (1) to strongly agree (6).

a. Positive Self-Evaluation

This dimension reflects positive evaluations of oneself, including confidence in personal abilities, qualities, and overall self-worth [8], [9].

i. Overall, I feel satisfied with myself [8], [9].
ii. I feel that I have a number of good qualities [8], [9].
iii. I feel that I am a valuable person [8],[9].
iv. I have abilities as good as other people [8], [9].

b. Negative Self-Evaluation

This dimension reflects tendencies to evaluate oneself negatively, including self-doubt, feelings of inadequacy, and negative perceptions of personal worth [8], [9].

i. I feel I have a few failures [8], [9].
ii. I rarely feel useless [8], [9].
iii. I don't feel less compared to other people [8],[9].
iv. I am someone who rarely thinks negatively [8], [9].

c. Self-Acceptance

This dimension reflects an individual's ability to accept and appreciate both their strengths and weaknesses while maintaining a positive attitude toward oneself, realistically and without denial [8], [29].

i. I am not too harsh on criticizing myself when making a mistake [29].
ii. I wish I could respect myself more [29].
iii. I can accept my strengths and weaknesses as they are [8], [9], [29].
iv. I feel comfortable being myself without having to pretend [29].